PG No. 15 D_{4h} $4/mmm$ [tetragonal] (internal polar dipole)

* Harmonics for rank 0

$\vec{\mathbb{Q}}_0^{(1,1)}[q](A_{1g})$

** symmetry

1

** expression

$$\frac{\sqrt{3}Q_x x}{3} + \frac{\sqrt{3}Q_y y}{3} + \frac{\sqrt{3}Q_z z}{3}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_1^{(1,0)}[q](A_{2g})$

** symmetry

z

** expression

$$\frac{\sqrt{2}Q_x y}{2} - \frac{\sqrt{2}Q_y x}{2}$$

$\vec{\mathbb{G}}_{1,1}^{(1,0)}[q](E_g)$, $\vec{\mathbb{G}}_{1,2}^{(1,0)}[q](E_g)$

** symmetry

x

$-y$

** expression

$$\frac{\sqrt{2}Q_y z}{2} - \frac{\sqrt{2}Q_z y}{2}$$

$$\frac{\sqrt{2}Q_x z}{2} - \frac{\sqrt{2}Q_z x}{2}$$

$\vec{\mathbb{Q}}_1^{(1,-1)}[q](A_{2u})$

** symmetry

z

** expression

Q_z

$\vec{\mathbb{Q}}_1^{(1,1)}[q](A_{2u})$

** symmetry

z

** expression

$$\frac{3\sqrt{10}Q_x xz}{10} + \frac{3\sqrt{10}Q_y yz}{10} - \frac{\sqrt{10}Q_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{Q}}_{1,1}^{(1,-1)}[q](E_u)$, $\vec{\mathbb{Q}}_{1,2}^{(1,-1)}[q](E_u)$

** symmetry

x

y

** expression

Q_x

Q_y

$\vec{\mathbb{Q}}_{1,1}^{(1,1)}[q](E_u)$, $\vec{\mathbb{Q}}_{1,2}^{(1,1)}[q](E_u)$

** symmetry

x

y

** expression

$$\frac{\sqrt{10}Q_x\left(2x^2-y^2-z^2\right)}{10} + \frac{3\sqrt{10}Q_y xy}{10} + \frac{3\sqrt{10}Q_z xz}{10}$$

$$\frac{3\sqrt{10}Q_x xy}{10} - \frac{\sqrt{10}Q_y\left(x^2-2y^2+z^2\right)}{10} + \frac{3\sqrt{10}Q_z yz}{10}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](A_{1g})$

** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$-\frac{\sqrt{6}Q_x x}{6} - \frac{\sqrt{6}Q_y y}{6} + \frac{\sqrt{6}Q_z z}{3}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](A_{1g})$

** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$-\frac{\sqrt{21}Q_x x\left(x^2+y^2-4z^2\right)}{14} - \frac{\sqrt{21}Q_y y\left(x^2+y^2-4z^2\right)}{14} - \frac{\sqrt{21}Q_z z\left(3x^2+3y^2-2z^2\right)}{14}$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](B_{1g})$

** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{2}Q_x x}{2} - \frac{\sqrt{2}Q_y y}{2}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](B_{1g})$

** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{7}Q_x x\left(3x^2-7y^2-2z^2\right)}{14} + \frac{\sqrt{7}Q_y y\left(7x^2-3y^2+2z^2\right)}{14} + \frac{5\sqrt{7}Q_z z\left(x-y\right)\left(x+y\right)}{14}$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](B_{2g})$

** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{2}Q_x y}{2} + \frac{\sqrt{2}Q_y x}{2}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](B_{2g})$

** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{7}Q_x y\left(4x^2-y^2-z^2\right)}{7} - \frac{\sqrt{7}Q_y x\left(x^2-4y^2+z^2\right)}{7} + \frac{5\sqrt{7}Q_z xyz}{7}$$

$\vec{\mathbb{Q}}_{2,1}^{(1,-1)}[q](E_g)$, $\vec{\mathbb{Q}}_{2,2}^{(1,-1)}[q](E_g)$

** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

** expression

$$\frac{\sqrt{2}Q_y z}{2} + \frac{\sqrt{2}Q_z y}{2}$$

$$\frac{\sqrt{2}Q_x z}{2} + \frac{\sqrt{2}Q_z x}{2}$$

$\vec{\mathbb{Q}}^{(1,1)}_{2,1}[q](E_g)$, $\vec{\mathbb{Q}}^{(1,1)}_{2,2}[q](E_g)$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

** expression

$$\frac{5\sqrt{7}Q_x xyz}{7} - \frac{\sqrt{7}Q_y z\left(x^2 - 4y^2 + z^2\right)}{7} - \frac{\sqrt{7}Q_z y\left(x^2 + y^2 - 4z^2\right)}{7}$$

$$\frac{\sqrt{7}Q_x z\left(4x^2 - y^2 - z^2\right)}{7} + \frac{5\sqrt{7}Q_y xyz}{7} - \frac{\sqrt{7}Q_z x\left(x^2 + y^2 - 4z^2\right)}{7}$$

$\vec{\mathbb{G}}^{(1,0)}_{2}[q](A_{1u})$
** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{\sqrt{6}Q_x yz}{2} - \frac{\sqrt{6}Q_y xz}{2}$$

$\vec{\mathbb{G}}^{(1,0)}_{2}[q](B_{1u})$
** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{2}Q_x yz}{2} + \frac{\sqrt{2}Q_y xz}{2} - \sqrt{2}Q_z xy$$

$\vec{\mathbb{G}}^{(1,0)}_{2}[q](B_{2u})$
** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{2}Q_x xz}{2} + \frac{\sqrt{2}Q_y yz}{2} + \frac{\sqrt{2}Q_z\left(x - y\right)\left(x + y\right)}{2}$$

$\vec{\mathbb{G}}^{(1,0)}_{2,1}[q](E_u)$, $\vec{\mathbb{G}}^{(1,0)}_{2,2}[q](E_u)$
** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$\frac{\sqrt{2}Q_x\left(y - z\right)\left(y + z\right)}{2} - \frac{\sqrt{2}Q_y xy}{2} + \frac{\sqrt{2}Q_z xz}{2}$$

$$-\frac{\sqrt{2}Q_x xy}{2} + \frac{\sqrt{2}Q_y\left(x - z\right)\left(x + z\right)}{2} + \frac{\sqrt{2}Q_z yz}{2}$$

* Harmonics for rank 3
$\vec{\mathbb{G}}^{(1,0)}_{3}[q](A_{2g})$
** symmetry

$$-\frac{3x^2 z}{2} - \frac{3y^2 z}{2} + z^3$$

** expression

$$-\frac{\sqrt{3}Q_x y\left(x^2 + y^2 - 4z^2\right)}{4} + \frac{\sqrt{3}Q_y x\left(x^2 + y^2 - 4z^2\right)}{4}$$

$\vec{\mathbb{G}}^{(1,0)}_{3}[q](B_{1g})$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{5}Q_x x\left(y-z\right)\left(y+z\right)}{2}-\frac{\sqrt{5}Q_y y\left(x-z\right)\left(x+z\right)}{2}+\frac{\sqrt{5}Q_z z\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{G}}_3^{(1,0)}[q](B_{2g})$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{5}Q_x y\left(x^2-y^2+2z^2\right)}{4}-\frac{\sqrt{5}Q_y x\left(x^2-y^2-2z^2\right)}{4}-\sqrt{5}Q_z xyz$$

$\vec{\mathbb{G}}_{3,1}^{(1,0)}[q](E_g,1)$, $\vec{\mathbb{G}}_{3,2}^{(1,0)}[q](E_g,1)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$\frac{3x^2y}{2}-y^3+\frac{3yz^2}{2}$$

** expression

$$\frac{\sqrt{3}Q_y z\left(4x^2-y^2-z^2\right)}{4}-\frac{\sqrt{3}Q_z y\left(4x^2-y^2-z^2\right)}{4}$$

$$-\frac{\sqrt{3}Q_x z\left(x^2-4y^2+z^2\right)}{4}+\frac{\sqrt{3}Q_z x\left(x^2-4y^2+z^2\right)}{4}$$

$\vec{\mathbb{G}}_{3,1}^{(1,0)}[q](E_g,2)$, $\vec{\mathbb{G}}_{3,2}^{(1,0)}[q](E_g,2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

** expression

$$-\sqrt{5}Q_x xyz+\frac{\sqrt{5}Q_y z\left(2x^2+y^2-z^2\right)}{4}+\frac{\sqrt{5}Q_z y\left(2x^2-y^2+z^2\right)}{4}$$

$$\frac{\sqrt{5}Q_x z\left(x^2+2y^2-z^2\right)}{4}-\sqrt{5}Q_y xyz-\frac{\sqrt{5}Q_z x\left(x^2-2y^2-z^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](A_{2u})$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{15}Q_x xz}{5}-\frac{\sqrt{15}Q_y yz}{5}-\frac{\sqrt{15}Q_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](A_{2u})$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{5Q_x xz\left(3x^2+3y^2-4z^2\right)}{12}-\frac{5Q_y yz\left(3x^2+3y^2-4z^2\right)}{12}+\frac{Q_z\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](B_{1u})$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$Q_x yz + Q_y xz + Q_z xy$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](B_{1u})$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{15}Q_x yz\left(6x^2 - y^2 - z^2\right)}{6} - \frac{\sqrt{15}Q_y xz\left(x^2 - 6y^2 + z^2\right)}{6} - \frac{\sqrt{15}Q_z xy\left(x^2 + y^2 - 6z^2\right)}{6}$$

$\vec{\mathbb{Q}}_3^{(1,-1)}[q](B_{2u})$
** symmetry

$$\frac{\sqrt{15}x^2 z}{2} - \frac{\sqrt{15}y^2 z}{2}$$

** expression

$$Q_x xz - Q_y yz + \frac{Q_z\left(x - y\right)\left(x + y\right)}{2}$$

$\vec{\mathbb{Q}}_3^{(1,1)}[q](B_{2u})$
** symmetry

$$\frac{\sqrt{15}x^2 z}{2} - \frac{\sqrt{15}y^2 z}{2}$$

** expression

$$\frac{\sqrt{15}Q_x xz\left(5x^2 - 9y^2 - 2z^2\right)}{12} + \frac{\sqrt{15}Q_y yz\left(9x^2 - 5y^2 + 2z^2\right)}{12} - \frac{\sqrt{15}Q_z\left(x - y\right)\left(x + y\right)\left(x^2 + y^2 - 6z^2\right)}{12}$$

$\vec{\mathbb{Q}}_{3,1}^{(1,-1)}[q](E_u, 1)$, $\vec{\mathbb{Q}}_{3,2}^{(1,-1)}[q](E_u, 1)$
** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$-\frac{3x^2 y}{2} + y^3 - \frac{3yz^2}{2}$$

** expression

$$\frac{\sqrt{15}Q_x\left(2x^2 - y^2 - z^2\right)}{10} - \frac{\sqrt{15}Q_y xy}{5} - \frac{\sqrt{15}Q_z xz}{5}$$

$$-\frac{\sqrt{15}Q_x xy}{5} - \frac{\sqrt{15}Q_y\left(x^2 - 2y^2 + z^2\right)}{10} - \frac{\sqrt{15}Q_z yz}{5}$$

$\vec{\mathbb{Q}}_{3,1}^{(1,-1)}[q](E_u, 2)$, $\vec{\mathbb{Q}}_{3,2}^{(1,-1)}[q](E_u, 2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$\frac{\sqrt{15}x^2 y}{2} - \frac{\sqrt{15}yz^2}{2}$$

** expression

$$\frac{Q_x\left(y - z\right)\left(y + z\right)}{2} + Q_y xy - Q_z xz$$

$$Q_x xy + \frac{Q_y\left(x - z\right)\left(x + z\right)}{2} - Q_z yz$$

$\vec{\mathbb{Q}}_{3,1}^{(1,1)}[q](E_u, 1)$, $\vec{\mathbb{Q}}_{3,2}^{(1,1)}[q](E_u, 1)$
** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

** expression

$$\frac{Q_x\left(8x^4-24x^2y^2-24x^2z^2+3y^4+6y^2z^2+3z^4\right)}{12}+\frac{5Q_yxy\left(4x^2-3y^2-3z^2\right)}{12}+\frac{5Q_zxz\left(4x^2-3y^2-3z^2\right)}{12}$$

$$-\frac{5Q_xxy\left(3x^2-4y^2+3z^2\right)}{12}+\frac{Q_y\left(3x^4-24x^2y^2+6x^2z^2+8y^4-24y^2z^2+3z^4\right)}{12}-\frac{5Q_zyz\left(3x^2-4y^2+3z^2\right)}{12}$$

$\vec{\mathbb{Q}}^{(1,1)}_{3,1}[q](E_u,2)$, $\vec{\mathbb{Q}}^{(1,1)}_{3,2}[q](E_u,2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$\frac{\sqrt{15}x^2y}{2}-\frac{\sqrt{15}yz^2}{2}$$

** expression

$$\frac{\sqrt{15}Q_x\left(y-z\right)\left(y+z\right)\left(6x^2-y^2-z^2\right)}{12}-\frac{\sqrt{15}Q_yxy\left(2x^2-5y^2+9z^2\right)}{12}+\frac{\sqrt{15}Q_zxz\left(2x^2+9y^2-5z^2\right)}{12}$$

$$\frac{\sqrt{15}Q_xxy\left(5x^2-2y^2-9z^2\right)}{12}-\frac{\sqrt{15}Q_y\left(x-z\right)\left(x+z\right)\left(x^2-6y^2+z^2\right)}{12}+\frac{\sqrt{15}Q_zyz\left(9x^2+2y^2-5z^2\right)}{12}$$